// FINAL RESEARCH NOTE / EN

01

ORB + SHOCK REVERSAL NQ FUTURES

Opening-range breakout and intraday shock reversal. IS research and OOS evaluation with frozen parameters.

INSTRUMENT	TIMEFRAME	OOS SAMPLE	OOS PERIOD
NQ Futures	15 minutes	277 trades	2025 / 2026*

* OOS through 21 September 2026. Report dated 22 September 2026.

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ORB + Shock Reversal

Final research report | Institutional summary

Two intraday hypotheses, one coordinated framework

The setup combines an opening-range breakout with an afternoon contrarian module. Its objective is to increase trading frequency in Nasdaq-100 futures while preserving the original ORB and avoiding overlapping positions.

Sample	Selection	Status
IS 2017-2024	100,000 configurations	Optimization and selection
OOS 2025-21/09/2026	One frozen candidate	No OOS optimization

Combined OOS: 277 trades | USD 26,597 net profit | 26.60% on a USD 100,000 reference account. The added module contributes 83 trades and USD 9,533.

Research conclusion

The candidate meets the aggregate numerical OOS criteria fixed before the test. Statistical robustness remains unconfirmed: bootstrap intervals include negative contributions and combined drawdown increases. OOS was explicitly authorized as an exploratory test after the IS bootstrap gate failed.

Scope: original ORB + Shock Reversal only. Discarded variants are omitted. The description preserves the economic and operational rationale without disclosing every threshold or implementation detail.

01 / Hypotheses and structure

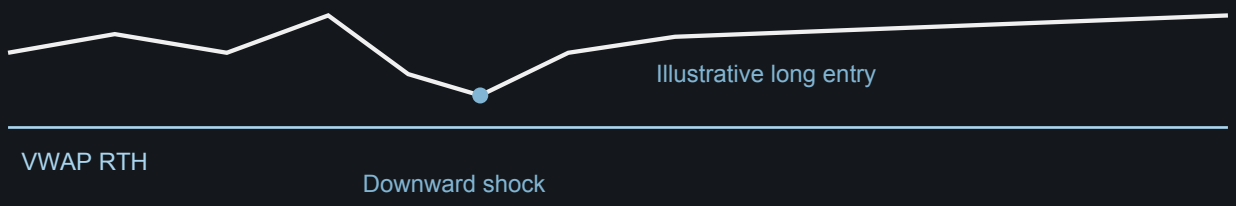
Temporal complementarity and exposure control

A. ORB: continuation after the open

The first RTH range summarizes the initial balance of buying and selling pressure. A completed-bar breakout consistent with overnight direction and an eligible volatility regime is treated as a continuation signal. Original ORB parameters are retained without recalibration in this study.

B. Shock Reversal: partial reversal of a local shock

The second module identifies intraday moves that are unusually large relative to the same time of day. The hypothesis is that transient price pressure may partially reverse. The signal opposes the local shock, while the VWAP condition requires alignment with the price location within the session.



Conceptual sketch: a local decline remains above VWAP. This is not an actual trade.

Reference: Zawadowski, Andor and Kertész, Short-term market reaction after extreme price changes of liquid stocks [1]. The paper examines liquid stocks and intraday shocks. The NQ application, afternoon timing, filters and risk rules are adaptations, not a replication of the academic result.

02 / Trading logic

Functional description with selective disclosure

Stage	Function
Context	15-minute NQ, overnight history, New York times.
ORB priority	The original module has priority; no overlapping positions.
Event	30-minute shock; absolute and relative comparison with the same time slot over 60 prior sessions.
Confirmation	ADX, ATR regime and position relative to RTH VWAP. RVOL was tested but is disabled in the candidate.
Execution	Completed-bar signal, next-bar-open entry. At most one Shock trade per day.
Risk	Module-specific stop and target. No position scaling or averaging down.

Daily sequence

09:30-09:45 NY: opening range → ORB through 13:00 → Shock between 14:00 and 15:00 → flat by 16:00.

A setup must occur on a later bar after the ORB exit; re-entry on the exit bar is disallowed. A downward shock permits a long only above VWAP; an upward shock permits a short only below VWAP. Both sides are symmetric.

The added module has an eight-bar holding cap, always bounded by the RTH close. At the selected entry times, the 16:00 limit occurs first or simultaneously. Sizing remains fixed at one NQ and does not depend on equity.

03 / Protocol and verification

IS selection separated from OOS evaluation

IS 2017-2024. 100,000 unique configurations of the added module were evaluated: stop/target, shock magnitude and duration, entry timing, holding period and ADX, ATR, VWAP and RVOL filters. ORB remained frozen.

Selection required combined and added average trade >USD 75, win rate >60% for both, and at least 192 added trades. Ranking prioritized frequency, then daily Sharpe and added average. Further checks required positive contributions in both IS subperiods and added average >USD 75 under two-tick stress.

Documented exception. IS bootstrap failed the predefined gate. The user authorized one exploratory OOS evaluation of the already frozen candidate; parameters were neither replaced nor changed.

OOS 01 Jan 2025-21 Sep 2026. No OOS optimization. Aggregate criteria: added and combined average >USD 75, combined win rate >60%, more trades than ORB and positive added profit under two-tick stress. The candidate meets these criteria.

Technical checks completed

Next-bar entry; ORB priority; conservative treatment of bars touching both stop and target; long/short symmetry; historical feature windows excluding future data. VectorBT PnL reconciliation: maximum difference USD 0 across 1,188 IS and 277 OOS trades. Python checks also covered the 60-session history and ADX translation. PowerLanguage has not yet been compiled in MultiCharts.

OOS history had already been observed in prior research: the split prevents recalibration in this test but does not make the history a fresh, independent validation.

04 / In-sample results

2017-2024 | Selection sample

Metric	ORB	ORB + Shock
Net profit	187,798 USD	218,533 USD
Return / account*	187.80%	218.53%
Profit factor	1.458	1.424
Win rate	61.14%	61.78%
Trade	898	1,188
Average trade	209.13 USD	183.95 USD
Max closed-trade DD	10,821 USD	13,973 USD



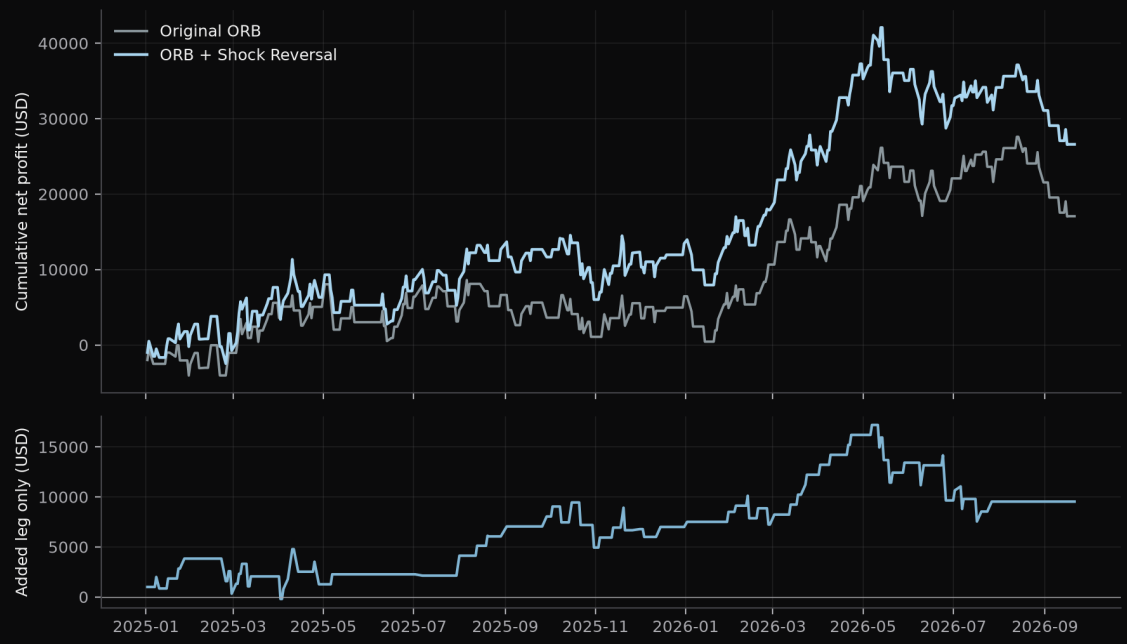
The module adds 290 trades (+32.3%), USD 30,735 and USD 105.98 average profit. Added profit factor is 1.294; win rate is 63.79%. Higher frequency increases profit but reduces combined average trade and profit factor relative to ORB.

*Reference notional account: USD 100,000. Return = net profit / 100,000; not annualized, not compounded and not a return on margin. Fixed one-NQ sizing; commissions included, zero slippage in the table.

05 / Out-of-sample results

2025-21 Sep 2026 | Unchanged parameters, no optimization

Metric	ORB	ORB + Shock
Net profit	17,064 USD	26,597 USD
Return / account*	17.06%	26.60%
Profit factor	1.117	1.139
Win rate	60.31%	63.54%
Trade	194	277
Average trade	87.96 USD	96.02 USD
Max closed-trade DD	10,565 USD	15,519 USD

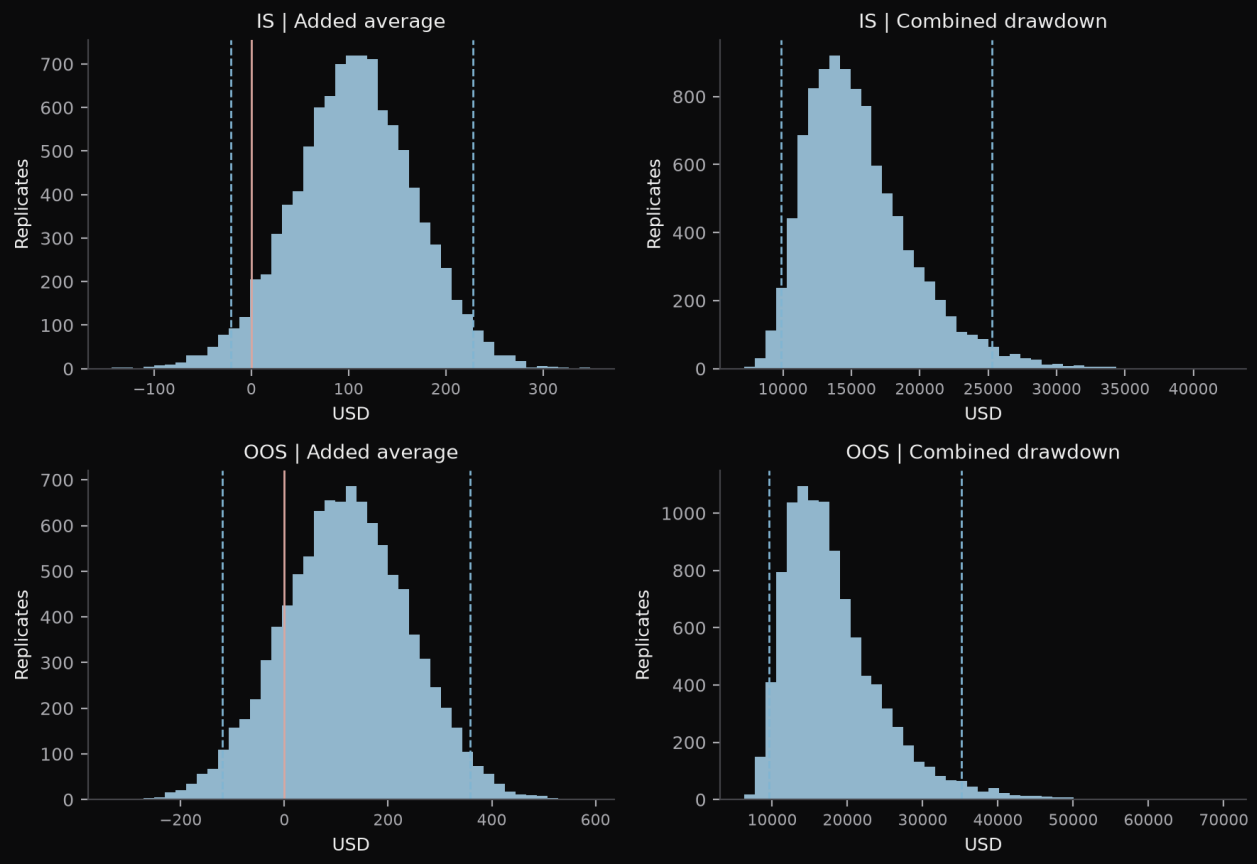


Added leg: 83 trades (+42.8%), USD 9,533, USD 114.86 average, 1.208 profit factor and 71.08% win rate. Combined max drawdown rises 46.9%. Contribution in 2025: +USD 6,997; partial 2026: +USD 2,536, with USD 70.44 average, below the USD 75 threshold in that individual year.

*Return uses the same USD 100,000 reference account, resetting cumulative profit at the start of OOS. It is not percentage growth from the final IS balance. Costs and sizing match the IS table.

06 / Bootstrap and costs

Distributions conditional on the selected candidate



Circular complete-session block bootstrap: 10,000 replicates for each block length 5/10/20 in IS; 10,000 ten-session-block replicates in OOS. ORB and the added module remain paired. Charts show ten-session blocks; dashed lines mark the 2.5th and 97.5th percentiles.

Added average, 95% interval	IS	OOS
USD	-20.75 / 227.84	-119.00 / 358.41

Both intervals include zero. Bootstrap does not repeat selection among 100,000 candidates and does not adjust for multiple testing. It does not estimate future profit probability. Resampled drawdowns refer to different IS and OOS horizons.

OOS stress: with two adverse ticks, combined profit is USD 24,462 (24.46% on the reference account); added contribution is USD 9,133 with USD 110.04 average. Commission: USD 4 per round trip. Targets fill at their limit; market and stop fills receive the stress.

07 / Essential and references

Conceptual code summary, not executable

```
ON BAR CLOSE:
  update context, volatility and RTH VWAP
  IF valid ORB AND position flat:
      schedule ORB entry at next open
  ELSE IF Shock window AND position flat:
      compare shock with same-time historical baseline
      IF anomaly AND filters agree AND quota available:
          direction = opposite to shock
          schedule entry at next open
          apply module-specific stop and target
  manage protective exits and close by end of RTH
  after a complete session: update historical memory
```

Operational thresholds, detailed filter calculations and state handling are intentionally omitted. This essential illustrates the architecture; it does not replace the complete PowerLanguage Signal. Data1: 15-minute NQ, full session; Data2 is not required for the selected RTH VWAP.

Definitions and interpretation

Net profit: sum of PnL after modeled commissions. Profit factor: sum of positive trade PnL divided by the absolute sum of negative trade PnL. Win rate: trades with positive net PnL / all trades. Max DD: largest peak-to-trough decline in closed-trade equity, not intrabar risk.

Sources and traceability

[1] Zawadowski, Andor & Kertész. Short-term market reaction after extreme price changes of liquid stocks. Working paper 2004; Quantitative Finance, 2006. arxiv.org/abs/cond-mat/0406696

[2] Local research dated 22 Sep 2026: protocol, SHOCK-015240 selection, trade ledgers, IS/OOS metrics, bootstrap and executed notebooks. Five-minute NQ data exported from MultiCharts and aggregated to 15 minutes; incomplete RTH sessions excluded.